Unit 1

Prerequisites

Knowledge of derivatives and integration

Introduction to Quantum Mechanics

Quantum mechanics is one of the most important intellectual achievement of the 20th century. This sophisticated field in physics has refined our understanding of the nature, particularly in the nanometer length scale down to atomic and subatomic level and has profound implications in chemistry, materials science, optics, and electronics. The existence of orbitals and energy levels in atoms can only be explained using quantum mechanics. It can aptly explain the behaviours of insulators, semi-conductors and conductors, as well as that of giant magneto-resistance. It can explain the quantization of light and both its corpuscular and wave nature.

Quantum mechanics has found myriad of applications in photonics, quantum electronics, and micro-electronics, nanotechnology and many more emerging technologies. Therefore, it is of extreme importance that scientists and engineers learn quantum mechanics.

Historical background

A series of experiments during the late nineteenth century, with puzzling and paradoxical results, proved impossible to be explained by the classical physics of mechanics and electromagnetism; and needed radically new concepts of theoretical physics. These experiments fell roughly into three categories:

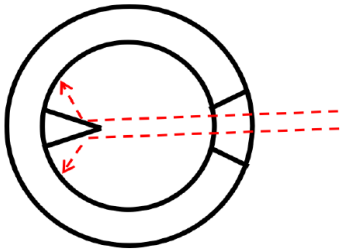
1. *The Corpuscular (particle-like) Behaviour of Light Waves*
   Black-body radiation, the photoelectric effect, the Compton effect.
2. *The Stability of the Atom*
   Why doesn't the electron fall into the nucleus as predicted by classical electromagnetic theory? What is the origin of atomic spectra?
3. *The Wave-like Behaviour of Particles*
   Electron diffraction

The emergence of quantum mechanics was an attempt to explain these phenomena and it has proved itself to be immensely accurate.

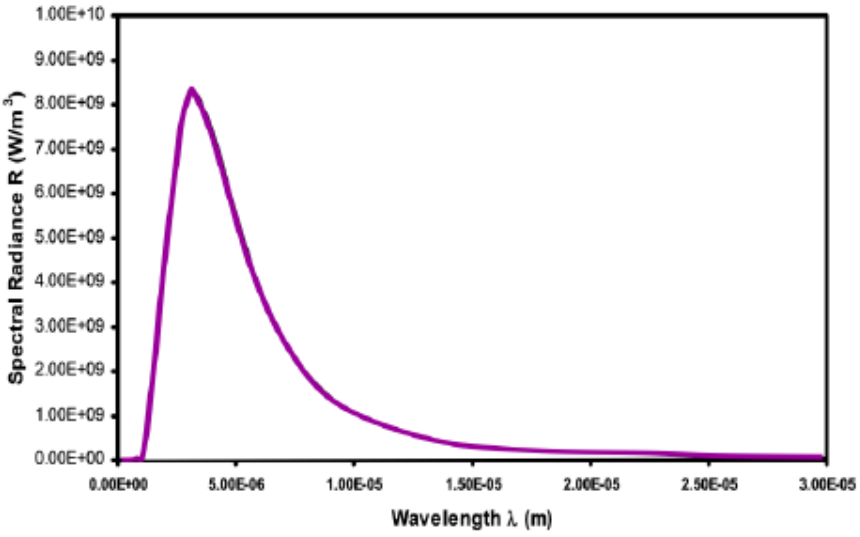
In the following sections we shall discuss these with considerable details.

Black-body radiation

All bodies give out radiation consistent with the temperature it is at. For example, at room temperature, humans give out infra-red (IR) radiation. This is the reason that militaries use IR goggles to spot people at night. At around 1000°C, bodies give out visible light, which is how conventional light bulbs function. When electromagnetic radiation is incident on a substance, some of it will be absorbed, some reflected and some transmitted. A substance can be imagined and constructed that absorbs all radiation incident on it as long as it cooler than its surroundings. This substance will also emit radiation as long as it is hotter than its surroundings. Such a substance is referred to as a "Black body". Graphite, as a material, comes close to satisfying this description. The figure below shows the schematic of the blackbody designed by Kirchoff. The arrows indicate that radiation entering the body will get absorbed by the internal surfaces of the body. Kirchhoff's law, states that the emissivity of a body which is in thermal equilibrium with its surrounding is equal to the absorptivity of the body.



In general, electromagnetic radiation emitted by a blackbody comes out over a range of wavelengths, however it is not emitted with uniform intensity across all wavelengths. The maximum intensity of the radiation occurs at one wavelength and the intensity decreases for all other wavelengths. The spectral distribution of the radiation emitted by blackbody is shown in figure below.



The spectral radiance plotted on the y-axis represents power per unit area per unit wavelength and is therefore represented by $W/m^2/m = W/m^3$, which is the unit shown on the y-axis. The area under the curve represents intensity (I), which is power per unit area.

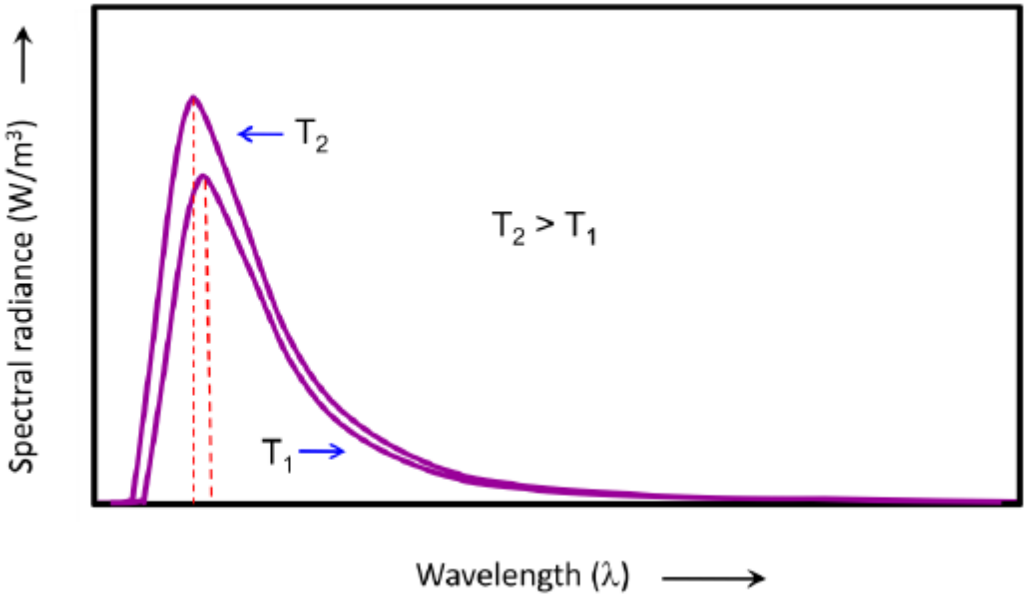
Mathematically, $I = \int_0^\infty R(\lambda)d\lambda$

There are two observations that can be made about blackbody radiation:

1) As temperature T of the body increases, intensity of the radiation from the body increases.

2) Higher the temperature, lower is the wavelength of the most intense part of the spectrum.

These two observations are represented by the figure below:



From this figure, it is evident that at the higher temperature T_2, the area under the curve, and hence intensity, has increased relative to the curve at T_1. Also, at the higher temperature T_2, the wavelength corresponding to the maximum intensity (identified using the red dotted lines in the figure), has decreased relative to that at T_1.

These two trends in blackbody radiation, can be mathematically stated in the form of two laws:

Stefan Boltzmann Law:

Josef Stefan based on experimental facts suggested that the total emissive power of a blackbody (I) is proportional to the fourth power of the absolute temperature.

$$I(T) = \sigma T^4$$

Here,

I is the total amount of radiation emitted by an object per square meter (Wm^{-2})
σ is a constant called the Stefan-Boltzmann constant (5.67 x 10-8 $Wm^{-2}K^{-4}$)
T is the absolute temperature of the body (in K)

Wein's displacement law

The variation of blackbody radiation with temperature shows that as the temperature increases the most intense part of the spectrum shifts towards the shorter wavelength. This is given by Wein's law

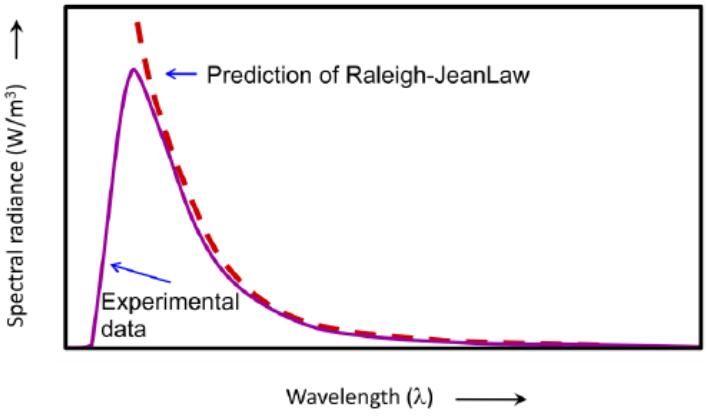
$$\lambda_{max}T = constant = 2898\ \mu mK$$

The scientific challenge that remained was to determine the exact form of the spectral radiance, or power per unit area at a particular wavelength, R (λ).

Several researchers worked to determine the form of R (λ). One of the early attempts, looked at the matter-radiation interaction in a classical manner, i.e. assumed an equipartition of energy, wherein all modes available to the solid through which it could absorb energy, participated in the process equally. This led to the law known as the Rayleigh-Jean law, which provides an equation for the spectral radiance as follows:

$$R(\lambda, T) = \frac{8\pi kT}{\lambda^4}$$

At higher values of λ, this led to a good match between theory and experiment. However as λ decreases, the theory predicts an ever increasing spectral radiance – a prospect dubbed as "Ultraviolet catastrophe". Common experience shows that this does not occur – bodies do not spontaneously release infinite energy. Therefore the Rayleigh-Jean law comprehensively fails at lower wavelengths. The mismatch between theory and experiment is shown in the schematic in figure below.



Hence, classical wave theory of radiation is not able to predict blackbody radiation. In order to explain these facts Max Planck (1901) gave a theory called Planck's quantum theory of radiation.The various postulates of quantum theory of radiations are as follows:

A radiation has energy. As light and heat are radiations, they are also associated with energy.

Radiant energy is not emitted or absorbed continuously but discontinuously in the form of small packets called photons. Photon is not a material body but is considered to be a massless packet of energy.

The energy E of a photon is related to the frequency of radiation, ν, by the equation $E = h\nu$, where h is Planck's constant.

Whenever a body emits or absorbs energy, it does so in whole number multiples of photons, i.e., $E = nh\nu$, where $n = 1, 2, 3$ etc.

The photoelectric effect

A series of experiments performed in the late nineteenth century revealed that electrons are emitted from a metal surface when light of sufficiently large frequency (excepting alkali metals all other metals require ultraviolet rays) falls upon it. This phenomenon is known as the *photoelectric effect*.

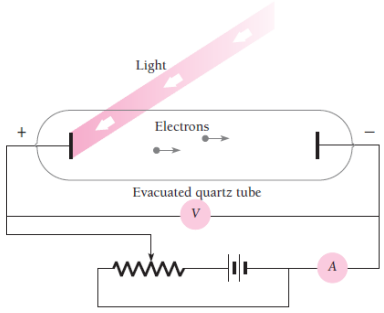


Fig. 1 Experimental set-up for photo-electric effect

The experimental setup is shown in fig. 1. The frequency υ and intensity I of the light, the material of the emitter and the retarding potential V can be varied. Some of the photoelectrons that emerge from the irradiated surface have sufficient energy to reach the cathode despite of its negative polarity and constitute the photo current that is measured by the ammeter in the circuit. In order to reach the collector the electrons must have a kinetic energy which is equal to more than the electrical potential energy that they must gain in going between emitter and collector, i.e.

$$\frac{1}{2}mv^2 \geq eV_0 \qquad (1)$$

V_0being the threshold potential that is needed to stop the electrons from constituting the photocurrent, which is also known as the stopping potential.

The experimental results can be summarized as

1. *It is an instantaneous process*. The time lag between the incidence of light and emission of electrons is ~ 10^{-9} s.
2. *There is a threshold frequency for the emission of photoelectrons*. If the light beam contains no photons having frequencies higher than this threshold, no photoelectron will be emitted (in other words, there will be no photo current) *regardless of the intensity of the light source*.
3. *The threshold frequency varies with different metals*.

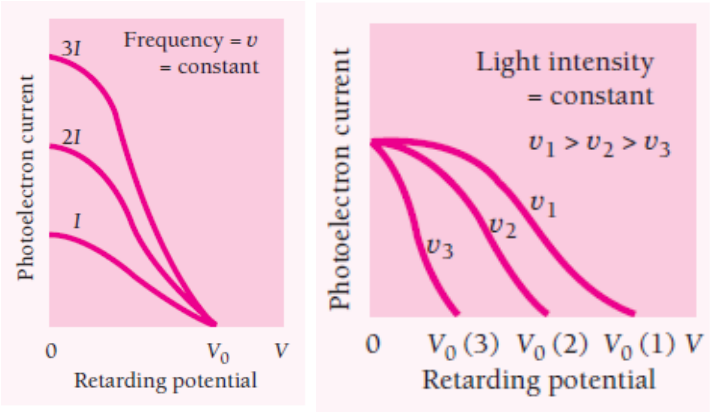


Fig. 2. Retarding potential vs. photocurrent graph at (a) constant frequency and (b) constant light intensity.

4. *The magnitude of the photocurrent is independent of the frequency of the light but is directly proportional to the intensity of the light. However, the average electron energy remains same* (Fig. 2a).
5. *For a given material, the higher the frequency of the light, the more energy the photoelectrons have and the value of stopping potential also increases* (Fig. 2b).

Failure of Classical electromagnetic theory

The existence of photoelectric effect might not be surprising, after all, light waves carry energy and some of the energy absorbed by the metal may somehow concentrate on individual electron and reappear as kinetic energy. But to explain the observed phenomena the classical electromagnetic theory fails to explain the following points.

1. The energy distributions of photo electrons is independent of the intensity of the light. A strong light beam yields more photoelectrons than a weak one of the same frequency, but the average electron energy is same. The electromagnetic theory of light, on the other hand, predicts that more intense the light, the greater the energies of the electrons.
2. That the photoelectron emission is almost an instantaneous process within the experimental accuracy (~10^{-9} s) is something that cannot be explained by classical theory, where it expects much more time lag.
3. A faint blue light produces more photoelectrons than those produced by bright red light.
4. Above the critical frequency, which varies with each metal, photoelectrons has an energy range from zero to a certain maximum value that increases linearly with increasing frequency (Fig. 3). This observation cannot be explained classically which predicts maximum kinetic energy should increase with the intensity of the light.

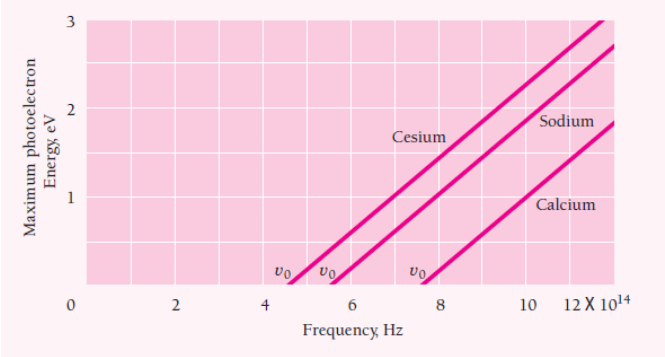


Fig. 3. Maximum photoelectron energy vs. frequency graph for the photo-electric effect.

Theory of photoelectric effect due to Einstein

The mystery of photoelectric effect was explained by Einstein in 1905 for which he was awarded the Nobel Prize in 1921. Based on Max Planck's suggestion that the radiation is emitted discontinuously as little bursts of energy called quanta, Einstein found that the quanta associated with a particular frequency υ of light all have same energy E, directly proportional to υ, i.e.

$$E = h\upsilon \qquad (2)$$

where, h is the Planck's constant (= 6.63 x 10^{-34}Js). He wrote the energy conservation law for photons knocking out electrons from a metal as

$$\frac{1}{2}mv^2 = h\upsilon - W \qquad (3)$$

Where m is the electron mass and W is called the work function and corresponds to the minimum energy required to knockout an electron from the metal surface. The functional relationship corresponding to Eqn. 3 is shown in Fig. 4.

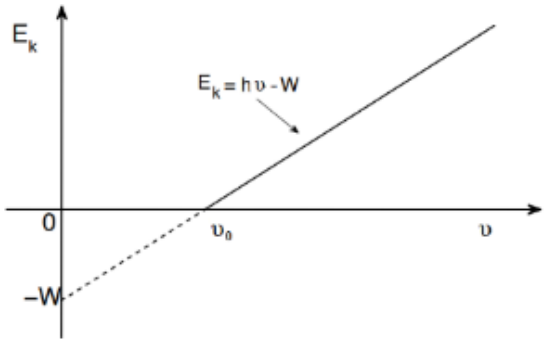


Fig. 4. Kinetic energy vs. frequency of the incident light graph

The concept of threshold frequency υ_0 can be easily understood from Eqn. 3 as the photoelectric effect can be observed *only* if

$$h\upsilon > W \tag{4}$$

Therefore, the threshold frequency is given by

$$\upsilon_0 = \frac{W}{h} \tag{5}$$

Where it is assumed that all of the photon energy is required to remove the electron and no energy is left over to provide its kinetic energy. Thus is explains why the threshold of electron emissions and the velocity of the emitted electrons depend on the frequency rather than the intensity of the light.

The Compton Effect

When short wavelength x-rays(photon) is incident on an element of low atomic weight material like carbon (for example, graphite),there is a decrease in energy of the photon (or, in other words, increase in wavelength)as a part of energy its initial energyis transferred to the electrons of the material which is scattered.This *effect* is called the *Compton Effect* or *Compton Scattering*.

According to the classical electromagnetism, the charged electron will begin to oscillate at the frequency of the incident wave, and will therefore radiate further electromagnetic waves at exactly the same frequency as the incident wave in all directions. This may be true for relatively longer wavelengths, but for experiments involving shorter wavelength X-rays incident on free electrons this is not the case. Compton explained this effect in terms of the scattering by electrons of individual photons.

Compton assumed the scattering of electromagnetic radiation from a charged particle as perfectly elastic, billiard ball type of collision between the photon and effectively free charged particle, as shown below in Fig. 1.

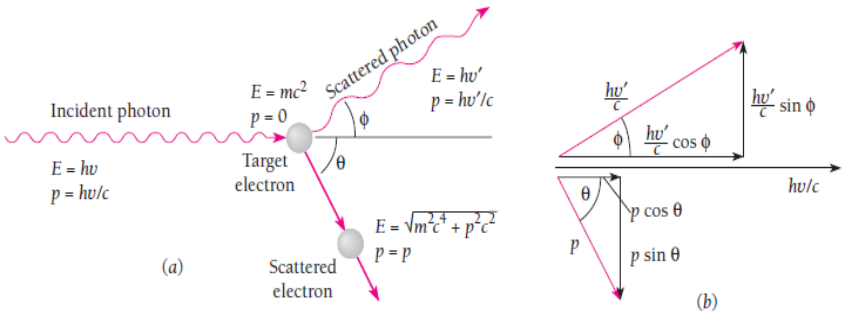


Fig. 1. Schematic for Compton Effect.

If the initial photon frequency is υ and the scattered photon has the lower frequency of υ', then

Loss of photon energy = gain in electron energy

$$h\upsilon - h\upsilon' = \text{KE} \qquad (1)$$

Since the photon rest mass is zero, its momentum p is related to its energy E by

$$p = \frac{E}{c} = \frac{h\upsilon}{c} \qquad (2)$$

Therefore, the initial and scattered photon momenta are $h\upsilon/c$ and $h\upsilon'$, respectively. Whereas the initial and final electron momenta are respectively 0 and p. As momentum is a vector quantity, in the collision it must be conserved in each of two mutually perpendicular direction.

In the original photon direction, conservation of momentum can be written as

$$\frac{h\upsilon}{c} + 0 = \frac{h\upsilon'}{c}cos\phi + p\,cos\theta \qquad (3)$$

And perpendicular to this direction

$$0 = \frac{h\upsilon'}{c}sin\phi - p\,sin\theta \qquad (4)$$

Here the angle ϕ is that between the directions of the initial and scattered photons, and θ is that between the initial photon and recoil electron.

Now multiplying Eqns. (3) and (4) by c and rearranging them we get

$$pccos\theta = h\upsilon - h\upsilon' cos\phi$$

$$pc\sin\theta = h\upsilon'\sin\phi$$

After squaring each of these equations and adding the new ones together we get

$$p^2c^2 = (h\upsilon)^2 - 2(h\upsilon)(h\upsilon')cos\phi + (h\upsilon')^2 \quad (5)$$

We now use the two relativistic energy equations of a particle

$$E = \text{KE} + mc^2 \quad (6)$$

$$E = (m^2c^4 + p^2c^2)^2 \quad (7)$$

Therefore

$$(\text{KE} + mc^2)^2 = m^2c^4 + p^2c^2$$

$$p^2c^2 = \text{KE}^2 + 2mc^2\,\text{KE} \quad (8)$$

Substitution of Eqn. (1) in Eqn. (8) gives

$$p^2c^2 = (h\upsilon)^2 - 2(h\upsilon)(h\upsilon') + (h\upsilon')^2 + 2mc^2(h\upsilon - h\upsilon')^2 \quad (9)$$

Putting this value of p^2c^2 in Eqn. (5) we obtain

$$2mc^2(h\upsilon - h\upsilon') = 2(h\upsilon)(h\upsilon')(1 - cos\phi) \quad (10)$$

Dividing Eqn. (10) by $2h^2c^2$, we get

$$\frac{mc}{h}\left(\frac{\upsilon}{c} - \frac{\upsilon'}{c}\right) = \frac{\upsilon}{c}\frac{\upsilon'}{c}(1 - cos\phi)$$

$$\frac{mc}{h}\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{(1 - cos\phi)}{\lambda\lambda'}$$

Rearranging we obtain

$$\lambda' - \lambda = \frac{h}{mc}(1 - cos\phi) \quad (11)$$

The quantity *h/mc* is known as *Compton wavelength* (λ_c); its value for an electron is 0.0243 Å. In terms of λ_c, Eqn. (11) becomes

$$\lambda' - \lambda = \lambda_c(1 - cos\phi) \quad (12)$$

The phenomenon described by Eqn. (12) is known as Compton effect and it strongly supports the quantum theory of radiation.

de Broglie's Hypothesis

From the study of black body radiation, photo electric effect and Compton effect it was necessary to assign particle like properties to radiation. The amount of energy associated with the radiant particle or photon is

$$E = h\upsilon = \frac{hc}{\lambda} \qquad (1)$$

where h, c, υ and λ are Planck's constant, speed of light, frequency and wavelength, respectively, and since $\upsilon\lambda = c$. Eqn. (1) proves that the radiation has particle like nature described by photons and, at the same time, wave like nature described by wavelength λ.

Louis de Broglie asserted that this dual property is not merely confined to photon or radiation, it can be attributed to any material particle. For his hypothesis de Broglie relied on the special theory of relativity and old quantum theory. These waves that arise out of matter particles are known as matter waves.

The energy E and momentum p of the photon is related by

$$p = \frac{E}{c} = \frac{h\upsilon}{c} = \frac{h}{\lambda}$$

The wavelength of a photon is therefore specified by its momentum according to the relation

$$\lambda = \frac{h}{p} \qquad (2)$$

de Broglie asserted that the nature is symmetric. Therefore, what is applicable to photon can also be applied to any other particle.

For a particle of mass *m* moving with a velocity *v*, the momentum is given by $p = mv$. So the wavelength associated with the material particle is

$$\lambda = \frac{h}{mv} \qquad (3)$$

Where m = relativistic mass$= \frac{m_0}{\sqrt{(1-\frac{v^2}{c^2})}}$, m_0 being the rest mass.

λ in Eqn. (3) is called de Broglie wavelength.

Experiments demonstrating wave properties of electron.

1. Davisson-Germer Experiment (*Experimental verification of de Broglie hypothesis*)

Clinton Davisson and Lester Germer in the USA and G.P. Thompson in the UK independently confirmed de Broglie's hypothesis experimentally. Davisson and Germer studied the diffraction

of electron beam using nickel crystal. If the electrons are accelerated by a potential difference of V volts it gains a kinetic energy given by

$$\frac{1}{2}mv^2 = eV \qquad (4)$$

Where v is the velocity attained by the electrons and $v << c$, c being the speed of light. The de Broglie wavelength of the electron is

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{(2meV)}} \qquad (5)$$

Here we put the value of v from Eqn. 4. Thus we get λ in terms of voltage V.

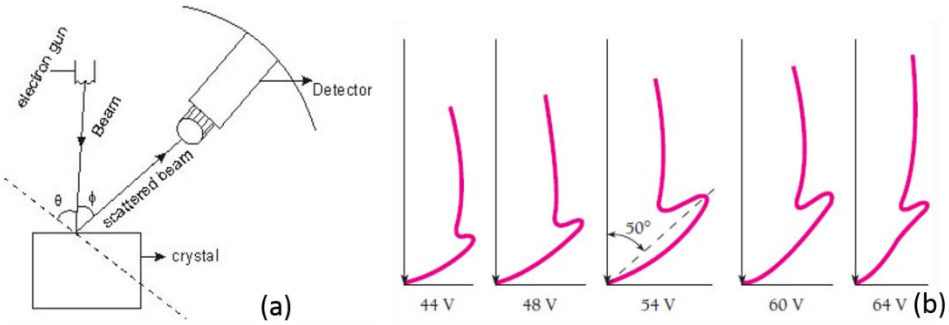


Fig. 1.(a) Experimental set-up for Davisson and Germer experiment and (b) its observation.

The experimental arrangement is shown in Fig. 1a. The electrons from a hot tungsten electron gun are collimated and incident normally on the crystal surface. The intensities of the electron beam scattered in all directions by the atoms of the crystal, which are arranged in a regular periodic fashion, were measured as a function of the angle of scattering. The results are shown in polar diagram (Fig.1b). The length of the radius vector is proportional to the intensity of the scattered beam and the angle of the radius vector and the Y-axis is the angle of scattering. Experimental results shows that the length of the peak increases until it reaches a maximum at $\phi = 50^o$ for $V = 54$ V. This can be explained as follows.

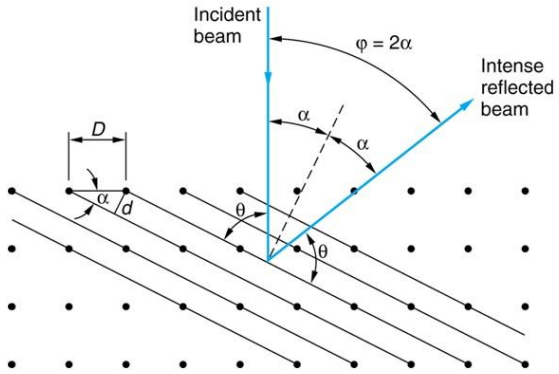


Fig. Schematic representations of atomic planes for Davison and Germer experiment.

The planes are shown in Fig. 2, where D is the distance between the consecutive atoms and d is the inter-planar separation. The Bragg's condition for constructive interference is

$$n\lambda = 2d\ sin\theta = 2d\ sin\ (90^o\text{-}\alpha) = 2d\ cos\ \alpha \qquad (6)$$

Again, from Fig. .

$$d = Dsin\ \alpha \qquad (7)$$

Combining Eqns. 6 and 7, we get

$$n\lambda = 2D\ sin\ \alpha.\cos\ \alpha = D\ sin\ 2\alpha$$

In terms of ϕ

$$n\lambda = D\ sin\ \phi \qquad (8)$$

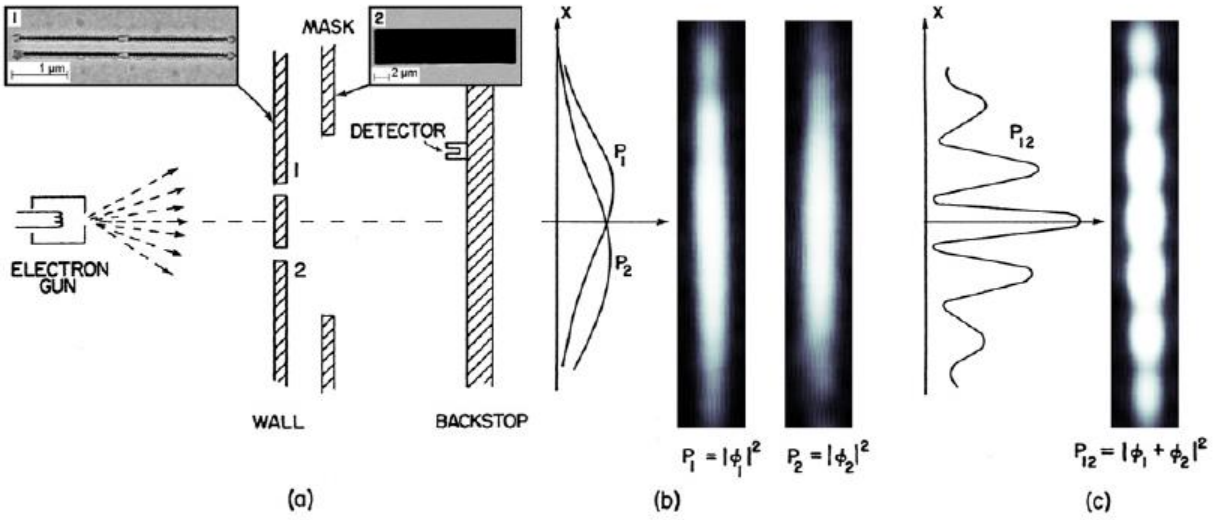
From x-ray data it is known that for nickel D = 2.15 Å. Therefore, for 54 V electron at ϕ= 50° and in the first order (i.e. n = 1)

$$\lambda = 2.15\ Å.\ Sin\ 50^o = 1.65\ Å$$

Further, with V =54 V and standard values of the constants Eqn. 5 can be used to obtain the wavelength of electron as λ = 1.66 Å.The remarkably close agreement between the two results proves the validity of de Broglie hypothesis.

2. Electron interference (double slit experiment)

Richard Feynman described electron diffraction as a phenomenon 'which has in it the heart of quantum mechanics. He went on to describe a thought experiment to help illustrate the phenomena of wave–particle duality, which is a postulate that all particles exhibit both wave and particle properties. Feynman's thought experiment is summarized in the figure below. Sending particles through a double-slit apparatus one at a time results in single particles appearing on the screen. Remarkably, however, an interference pattern emerges when these particles are allowed to build up one by one. The probability distributions in the three scenarios: electrons traveling through slit 1 with slit 2 closed (P1); electrons traveling through slit 2 with slit 1 closed (P2); and electrons traveling through both slits (P12), confirms the quantum mechanical superposition principle, i.e. the wave properties of electrons.



https://www.youtube.com/watch?v=ZqS8Jjkk1HI

HEISENBERG'S UNCERTAINTY PRINCIPLE

A moving particle is associated with de Broglie wave group. So it is better to represent it by a group of waves rather than by a localized entity. So there is a fundamental limit to the accuracy with which we can measure the properties of the particle, such as its momentum and position. When the de Broglie wave group is very narrow the position of the particle can be easily estimated within the width of the group, but the wavelength is very hard to establish (Fig. 1). If the wide of the group is wide, then wavelength can be easily established, but the particle may be anywhere within the wide width of the group (Fig. 2).

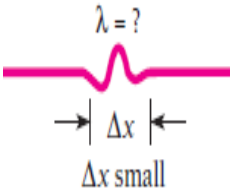


Fig. 1

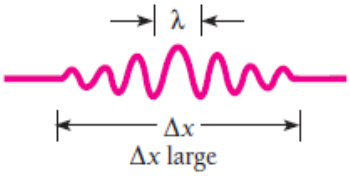


Fig. 2

Thus there is always an uncertainty in the simultaneous measurement of the momentum and position of the particle. The uncertainty principle states that 'The product of the uncertainty δx in the position of the body at some instant and uncertainty δp in its momentum at the same instant is equal to or greater than $h/2\pi$, i.e. the reduced Planck's constant ($\hbar$), h being Planck's constant'. i.e.

$$\delta x \, \delta p \geq h/2\pi$$

Now, let us consider two wave trains of slightly different angular frequency ω and propagation constant k are superimposed to form a group of waves

$$\Psi_1 = A \cos(\omega t - kx)$$

and $$\psi_2 = A \cos[(\omega + \delta\omega)t - (k + \delta k)x]$$

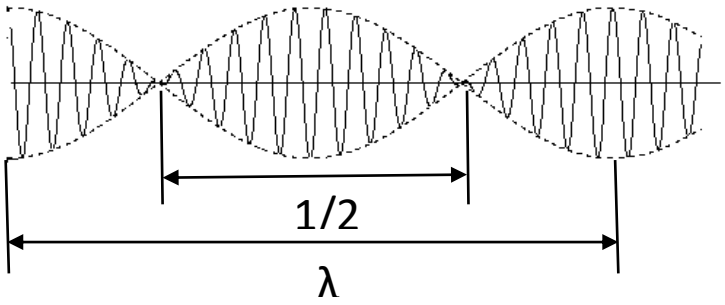
Superposition of these two waves yields

$$\Psi = \psi_1 + \psi_2 = 2A \cos\left[\frac{1}{2}(2\omega + \delta\omega)t - \frac{1}{2}(2k + \delta k)x\right] \cos\frac{1}{2}(\delta\omega t - \delta k x)$$

Since $\omega >> \delta\omega$ and $k >> \delta k$

$$\Psi = 2A \cos(\omega t - kx) \cos\left(\frac{\delta\omega}{2}t - \frac{\delta k}{2}x\right)$$

The width of each group is equal to half the wavelength λ_m of the modulation.



So we assume that the uncertainty δx is equal to width of the each group. i.e. $\delta x \cong \frac{1}{2}\lambda_m$, where $\lambda_m = 2\pi/k_m$.

The propagation constant of the modulation is $k_m = \delta k/2$

Therefore, $$\lambda_m = \frac{4\pi}{\delta k}$$

i.e. $$\delta x \cong \frac{1}{2} . \frac{4\pi}{\delta k} \cong \frac{2\pi}{\delta k}$$

or $$\delta k \cong \frac{2\pi}{\delta x} \qquad (1)$$

The de Broglie wavelength of a particle of momentum p is $\lambda = h/p$

And propagation constant

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/P} = \frac{2\pi p}{h} \qquad (2)$$

Uncertainty δk in the de Broglie wave results in the uncertainty of the particles momentum.

From Eqn. 2 we can write $p = \frac{h}{2\pi}k$

Therefore, $\delta p = \frac{h}{2\pi}\delta k \cong \frac{h}{2\pi}\frac{2\pi}{\delta x} = \frac{h}{\delta x}$

Or $\boldsymbol{\delta p . \delta x \cong h}$

In general, $\delta p . \delta x \geq \hbar$

The inequality sign is due to the fact that the wave groups may have different shapes.

Energy–Time Uncertainty

Suppose we want to measure the energy E during some time interval δt in an atomic process. If the energy is in the form of electromagnetic waves, the short time interval will restrict the measurement of frequency v of the waves with accuracy. In counting the number of waves v, the uncertainty is one wave, i.e. $\delta v = 1/\delta t$.

Now, we know that $E = hv$.

Therefore the energy uncertainty, $\delta E = h\delta v = h\frac{1}{\delta t} = \frac{h}{\delta t}$

Thus $\boldsymbol{\delta E . \delta t = h}$

The more general expression, $\delta E . \delta t \geq \hbar$

Example 1

In atomic transition from excited state the time interval is of the order of 10^{-8} s. So energy uncertainty

$$\delta E = \frac{h}{\delta t} = \frac{6.64\ x\ 10^{-34}}{2\ x\ 3.14\ x\ 10^{-8}}\frac{Joule-sec}{sec} \cong 1.1\ x\ 10^{-26} Joules$$

Hence $\delta v = \frac{\delta E}{h} \cong \frac{1.1\ x\ 10^{-26}}{6.64\ x\ 10^{-34}}\frac{Joule}{Joule-sec} \cong 1.6\ x\ 10^{7} sec^{-1}$

Applications of Heisenberg's uncertainty principle

1 Non-Existence of Electron in the Nucleus

The radius of the nucleus of an atoms is of the order of 10^{-14} m. If an electron is confined within the nucleus, the uncertainty in its position must not be greater than 10^{-14} m. According to uncertainty principle for the lowest limit of accuracy

$$\Delta x \, \Delta P = \frac{h}{2\pi}$$

Where Δx is uncertainty in the position and ΔP is the uncertainty in the momentum.

$$\Delta P = \frac{h}{2\pi \Delta X} = \frac{6.625 \times 10^{-34}}{2 \times 3.14 \times 2 \times 10^{-14}} \qquad (\text{as } \Delta X = \text{diameter of nucleus})$$

$$\Delta P = 5.275 \times 10^{-21} \text{ kg m/sec}$$

This is the uncertainty in momentum of the electron. It means the momentum of the electron would not be less than ΔP, rather it could be comparable to ΔP. Thus

$$\Delta P = 5.275 \times 10^{-21} \text{ kg m/sec}$$

The kinetic energy of the electron can be obtained in terms of momentum as

$$T = \tfrac{1}{2} mv^2 = p^2 / 2m$$

$$= \frac{(5.275 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31}} \, J$$

$$= \frac{(5.275 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} \, eV$$

$$= 95.55 \times 10^6 eV$$

$$\approx 96 MeV$$

From the above result, it is clear that the electrons inside the nucleus may exist only when it possesses the energy of the order of 96 MeV. However, the maximum possible kinetic energy of an electron emitted by radioactive nuclei has been found about 4MeV. Hence, it is concluded that the electron cannot reside in the nucleus.

Radius of Bohr's First orbit

If Δx and ΔP be the uncertainties in determining the position and momentum of the electron in the first orbit, then from the uncertainty principle

$$\Delta x \Delta P \approx \hbar$$

or $\Delta P \approx \hbar/\Delta x$ (i)

The uncertainty in kinetic energy (K.E) of electron may be written as

$$\Delta T = \frac{(\Delta P)^2}{2m} \qquad [KE=T=\frac{P^2}{2m}] \tag{ii}$$

From Eqs. (i) and (ii), we have

$$\Delta T = \frac{1}{2m}\ [\frac{\hbar}{\Delta x}]^2$$

And the uncertainty in the potential energy of the same electron is given by

$$\Delta V = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{\Delta x} \qquad\qquad [V = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{x}\ ]$$

The uncertainty in the total energy of electron together with Ze as the nucleus charge

$$\Delta E = \Delta T + \Delta V$$

$$= \frac{\hbar^2}{2m(\Delta x)^2} - \frac{(Ze)^2 1}{\Delta x 4\pi\varepsilon_0}$$

The condition for this uncertainty in the energy to be minimum is

$$\frac{d(\Delta E)}{d(\Delta x)} = 0$$

$$\text{Or} - \frac{\hbar^2}{m(\Delta x)^3} + \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{(\Delta x)^2} = 0\,, \qquad \Delta x = \frac{\hbar^2(4\pi\varepsilon_0)}{mZe^2}$$

$$r = \Delta x = \frac{\hbar^2(4\pi\varepsilon_0)}{mZe^2}$$

$\mathbf{r} = \frac{\boldsymbol{\varepsilon_0 h^2}}{\boldsymbol{\pi m Z e^2}}$ This is the radius of first Bohr's orbit.

N.B.

Uncertainty principle is not significant in case of macro-bodies

The Heisenberg Principle is of no practical importance for heavy bodies where the de Broglie wavelength is negligibly small.

For example, let us take the case of a cricket ball in flight. The indeterminacy in the position of the ball is, say, 1 mm. we can determine the indeterminacy of velocity of the ball from uncertainty principle.

$$\Delta x \Delta p \approx h$$

$$\therefore \Delta x m \Delta v \approx h$$

$$\Delta v \approx \frac{h}{m\Delta x} = \frac{6.62 \times 10^{-34} J.s}{0.5kg \times 10^{-3} m} \cong 10^{-30} m/s.$$

The above inaccuracy is negligible and not detectable. It implies that the uncertainties are of no importance in case of macro bodies; and the position and velocity of a macro body can be simultaneously determined with a high degree of accuracy. As a result, macroscopic body follows a well defined trajectory.